Conversational AI Powered by Large Language Models Amplifies False Memories in Witness Interviews

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ABSTRACT

This study examines the impact of AI on human false memories — recollections of events that did not occur or deviate from actual occurrences. It explores false memory induction through suggestive questioning in Human-AI interactions, simulating crime witness interviews. Four conditions were tested: control, survey-based, pre-scripted chatbot, and generative chatbot using a large language model (LLM). Participants (N=200) watched a crime video, then interacted with their assigned AI interviewer or survey, answering questions including five misleading ones. False memories were assessed immediately and after one week. Results show the generative chatbot condition significantly increased false memory formation, inducing over 3 times more immediate false memories than the control and 1.7 times more than the survey method. 36.4% of users' responses to the generative chatbot were misled through the interaction. After one week, the number of false memories induced by generative chatbots remained constant. However, confidence in these false memories remained higher than the control after one week. Moderating factors were explored: users who were less familiar with chatbots but more familiar with AI technology, and more interested in crime investigations, were more susceptible to false memories. These findings highlight the potential risks of using advanced AI in sensitive contexts, like police interviews, emphasizing the need for ethical considerations.

Introduction

False memories, defined as recollections of events that did not occur or that significantly deviate from actual occurrences, have been the subject of extensive research in psychology. The study of false memories is crucial due to their potential to distort testimonies, compromise legal proceedings, and lead to flawed decision-making based on misinformation, underscoring the far-reaching consequences of this form of deception[1–6]. An early contributor to the field was Bartlett[7], who posited that memory is a reconstructive process susceptible to various influencing factors. Research has demonstrated that memory retrieval is not an exact reproduction of past events, but rather a constructive process shaped by individual attitudes, expectations, and cultural contexts[8–10].

The research of Loftus and colleagues[5,6,11–13] have established false memories as a critical field of study in psychology. Their investigations into memory malleability and the misinformation effect have profoundly influenced the understanding of memory processes, with implications spanning psychology, law, and education[12–14]. Loftus and Palmer's seminal study[11] demonstrated the significant impact of question-wording on eyewitness memory. Participants viewing a car accident video provided markedly different speed estimates depending on the verb used in questioning (e.g., 'collided', 'bumped', 'contacted', or 'hit'). This finding revealed the susceptibility of memory to linguistic influence.

Further, the landmark "Lost in the Mall" experiment[6] demonstrated the possibility of implanting entirely false childhood memories. A recent replication by Murphy et al.[15] with a larger sample of participants showed that 35% of them reported a false memory of getting lost in a mall during childhood (compared to 25% in the original paper). These findings reinforce the robustness of the original study's conclusions and underscore the potential implications for eyewitness testimony in legal settings.

Neuroimaging studies have provided insights into the neural mechanisms of true and false memories. Slotnick and Schacter[2] used functional magnetic resonance imaging (fMRI) to investigate neural correlates of true and false recognition of abstract shapes, finding greater activation in early visual processing regions for true recognition. Gonsalves et al.[3] used event-related potentials (ERPs) to examine neural processes associated with false memory formation during encoding and retrieval. Stark et al. showed that false memories could be distinguished from true memories by sensory reactivation in the early regions of the

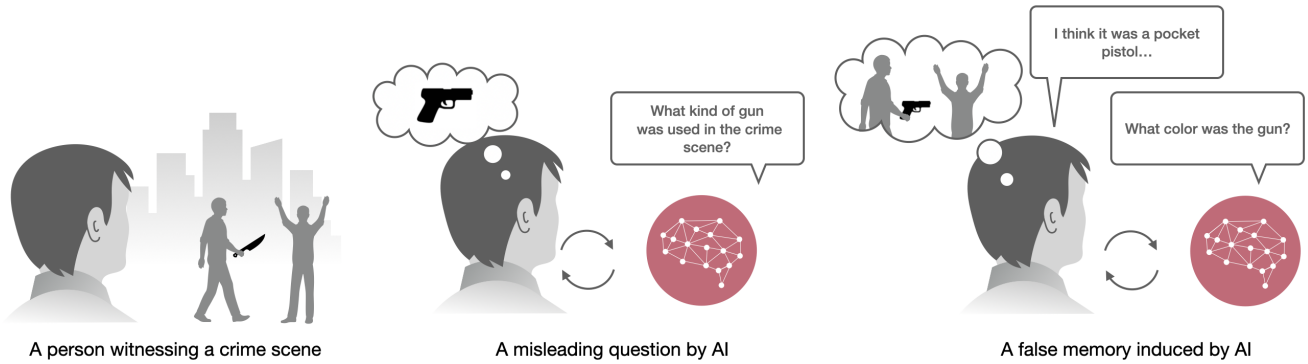

Figure 1. Manipulation of Eyewitness Memory by AI: This figure illustrates the process of AI-induced false memories in three stages. It begins with a person witnessing a crime scene involving a knife, then shows an AI system introducing misinformation by asking about a non-existent gun, and concludes with the witness developing a false memory of a gun at the scene. This sequence demonstrates how AI-guided questioning can distort human recall, potentially compromising the reliability of eyewitness testimony and highlighting the ethical concerns surrounding AI's influence on human memory and perception.

auditory and visual cortex[16]. Okado and Stark also found distinct patterns in the medial temporal lobe and prefrontal cortex regions when encoding false and true memories during the original event and misinformation phase[17]. Although neuroimaging can provide valuable insights into memory processes, the neural signals of true and false memories are very similar, the differences in neural reactivation and patterns are not sizable, and applying these techniques in real-world settings is not yet practical. The high cost, complex infrastructure requirements, and time-intensive nature of neuroimaging methods, such as fMRI, limit their applicability outside controlled laboratory environments.

A comprehensive mega-analysis by Scoboria et al.[18] addressed the variability in false memory estimates across studies by developing a standardized coding system by analyzing data from eight published implantation studies (N = 423). This systematic approach yielded a false memory formation rate of 30.4%, with an additional 23% of cases showing some degree of event acceptance. Notably, when the suggestion included self-relevant information, involved an imagination procedure, and was not accompanied by a photo depicting the event, the false memory formation rate increased to 46.1%. This analysis provides the most valid estimate of false memory formation and associated moderating factors such as age, cognitive abilities, and suggestibility, as well as the emotional valence of events and social pressure within the implantation literature to date.

These collective findings have reshaped our understanding of memory processes, emphasizing the dynamic and reconstructive nature of memory. The combination of behavioral studies, neuroimaging techniques, and large-scale analyses has provided a multifaceted view of false memory phenomena.

False Memories and Artificial Intelligence

In recent years, the rapid advancement of artificial intelligence (AI) technologies, particularly large language models[19] and visual models[20], has led to their widespread integration into work processes and daily life. From personal assistants[21] to virtual characters[22] and memory augmentation tools for the elderly[23], AI has become an integral part of human-computer interaction. However, this integration raises critical questions about the potential impact of AI on human cognition, particularly in the area of memory formation and retention.

A growing body of research has begun to explore the complex relationship between AI systems and human memory. Of particular concern is the dangerous potential for AI to contribute to the formation of false memories, as shown in figure 1. This concern is amplified by the known yet unresolved tendency of AI models to hallucinate or generate false information, either intentionally or unintentionally[24–27]. Initial studies have provided evidence for the potential of AI systems to influence memory formation. In a separate study, a social robot that provided users with incorrect information before a memory recognition test had an influence comparable to that of humans. The study found that even though the inaccurate information was emotionally neutral and not inherently memorable, 77% of the falsely provided words were incorporated into the participants' memories as errors[28].

While prior studies have largely examined how deepfakes and misleading information affect memory[29,30], the potential impact of conversations with a chatbot powered by an LLM on false memory formation remains an unexplored area. As these AI-driven dialogue systems become increasingly integrated into our daily lives, there is an urgent need to investigate their specific influence on the creation of false memories. This research gap is particularly significant given the rapid proliferation

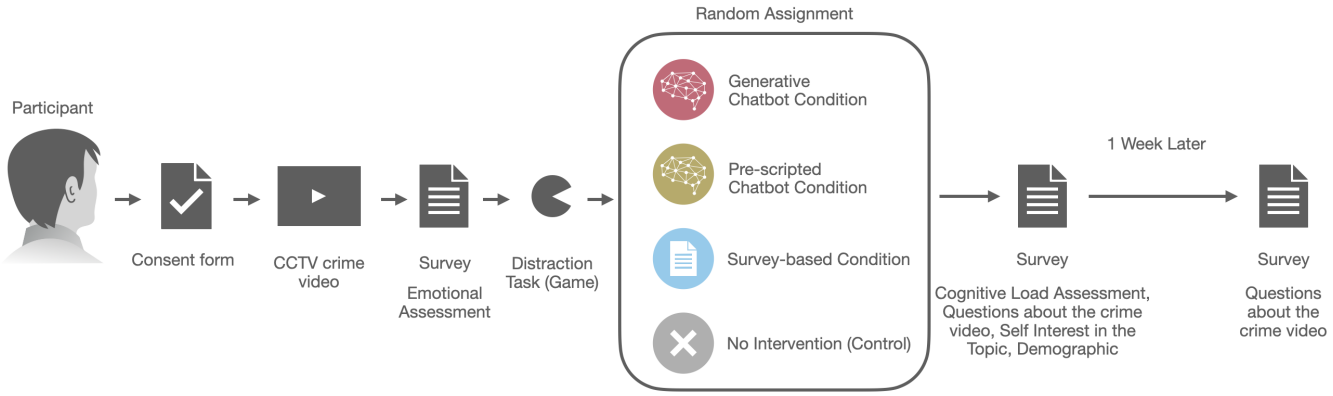

Figure 2. Experimental Design for Studying AI-Induced False Memories: This figure outlines a two-phase study on AI-induced false memories. In Phase 1, participants watch a CCTV crime video, complete emotional assessments and filler tasks, and are randomly assigned to one of four conditions: control, survey-based, pre-scripted chatbot, or generative chatbot. They then undergo cognitive load assessment and answer questions about the video. Phase 2, conducted one week later, involves participants recalling the video and answering the same questions, allowing researchers to measure the persistence of potential false memories induced by different AI interactions.

and adoption of conversational AI technologies across various industries and applications.

Research Questions and Hypotheses

To address this critical gap in our understanding, we conducted a comprehensive study investigating the impact of LLM-powered conversational AI on the formation of false memories. The study simulated a witness scenario where AI systems served as interrogators similar to Loftus's study[11], a situation we might encounter in future law enforcement or legal contexts.

This experimental design, as shown in Fig.2, involved 200 participants randomly assigned to one of four conditions in a two-phase study. We created a cover story to prevent people from figuring out the actual goal of the study; participants were told that this study seeks to evaluate what happens when people view video coverage of a crime.

In Phase 1, participants watched a two-and-a-half minute silent, non-pausable closed-circuit television (CCTV) video of an armed robbery at a store (Sayford Supermarket robbery on April 6, 2019), simulating a witness experience (Fig.3). They then interacted with their assigned condition, which was one of four experimental conditions designed to systematically compare various memory-influencing mechanisms:

- **Control Condition**: This condition serves as the baseline, where participants do not interact with any false memory-inducing method. After watching the video, participants in this condition proceed directly to the follow-up questions without any intervention.

- **Survey-based Condition**: In this condition, similar to the approach of Loftus (1975)[14], the participants complete a survey using Google Forms. The survey consists of 25 yes-or-no questions, five of which are misleading and serve as the focus of the study. The misleading questions are designed to induce false memories related to the video content. For example, one such question is: "Was there a security camera positioned in front of the store where the robbers dropped off the car?" In reality, this question is misleading because the robbers arrived on foot, not by car. The full list of questions is in the Supplementary Section.

- **Pre-scripted Chatbot Condition**: Participants were told that they were interacting with an AI police chatbot. They were asked to interact with a pre-scripted conversational agent that asked the same set of questions as the survey-based condition. The chatbot presents each question to the participant, waits for their response, and proceeds to the next question.

- **Generative Chatbot Condition**: Participants were told that they were interacting with an AI police chatbot and were asked to engage with a generative conversational agent that asked the same questions. However, the chatbot gives feedback to the participant's responses using an LLM. The chatbot was prompted to agree with the participant's answer and provide reinforcement, potentially strengthening the false memories. For instance, the chatbot asks a pre-scripted leading question containing false information implying the robbers arrived by car when they actually walked: "Was there a security camera positioned in front of the store where the robbers dropped off the car?" When the user incorrectly